

TITLE PAGE

- **Problem Statement ID** – 26126
- **Problem Statement Title**- Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor Environment
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 194940
- **Team Name** - PathRage



VisionNav: Camera-Only Autonomous Navigation for an Outdoor UGV

❖ Proposed Solution

Detailed explanation of the proposed solution

- **VisionNav**: camera-only autonomous navigation for an outdoor UGV – no GPS, no LiDAR
- One **stereo camera + IMU** feeds two parallel modules
- **Visual SLAM** – builds a 3D map and tracks vehicle pose
- **Perception AI** – depth-based obstacle detection + learned terrain segmentation
- Outputs fused into a **hierarchical global + local planner** (ROS 2 Nav2) that drives the UGV between waypoints

How it addresses the problem

Path detection → Perception AI

Separates safe ground from hazards in real time

Visual localization → Visual SLAM

Pose and heading estimation without GPS

Collision avoidance → Hierarchical planner

Reacts to sudden obstacles while still progressing to the goal

Innovation and uniqueness of the solution

- **Two-tier perception**: a geometric, depth-based safety layer underneath, a learned semantic layer on top
- **Stereo instead of monocular** – removes the scale-drift problem of pure monocular SLAM
- **Built on Nav2**, a mature planning stack, instead of a custom planner – lower implementation risk
- **Camera-only** design keeps hardware cost low

Technologies to be used

Hardware

- Stereo camera + IMU
- Onboard embedded GPU compute **[model]**

Visual SLAM

- ORB-SLAM3
- Robot_Localization (EKF)

Perception AI

- OpenCV, PyTorch
- Fast-SCNN / BiSeNetV2
- ONNX + TensorRT

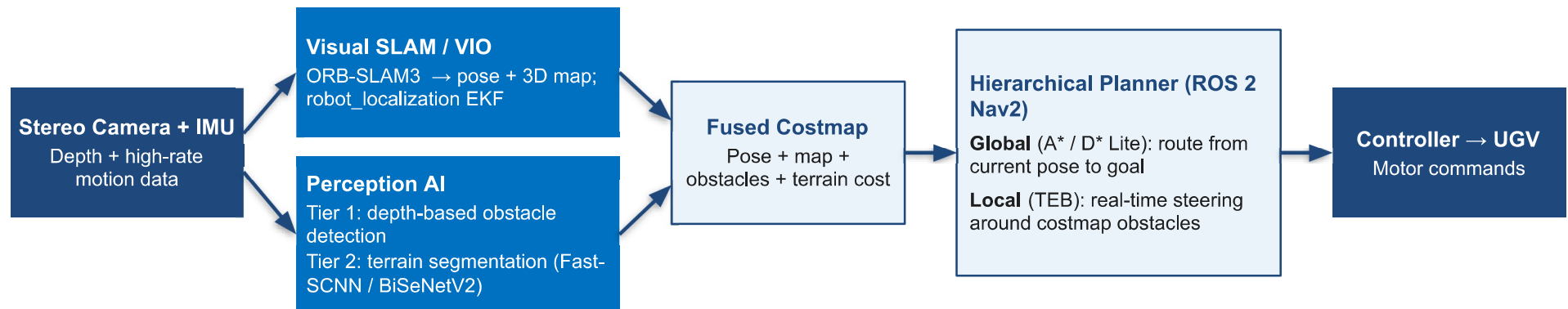
Navigation

- ROS 2 Humble, Nav2
- A* / D* Lite (global)
- TEB (local)

Simulation & Code

- Gazebo simulation
- Github:** <https://github.com/haricharan39/pathrage>

Methodology and process for implementation



Core idea – two-tier perception: Tier 1 is purely geometric (raw stereo depth, no learned model). Tier 2 is learned semantic segmentation. If Tier 2 is uncertain, the system falls back to Tier 1 – safety never depends on the AI being right.

Gazebo simulation on ROS 2 Humble validates the full pipeline before any hardware.

Analysis of the feasibility of the idea

- Built entirely from **mature, open-source components** – ROS 2, Nav2, ORB-SLAM3
- **No new algorithm** to invent – effort goes into integration and tuning
- Full pipeline validated in **Gazebo simulation** before any hardware deployment
- **Lightweight real-time models** (Fast-SCNN / BiSeNetV2) exported via ONNX / TensorRT for embedded compute
- **Status:** design and technology choices worked out; implementation, trials and benchmarks are the next phase
- **GitHub:** <https://github.com/haricharan39/pathrage>

Potential challenges and risks

1. Segmentation generalisation

Model may be uncertain on terrain unseen in training

2. Scale drift

Affects monocular-style vision systems

3. Changing outdoor lighting

Light varies constantly through the day

Strategies for overcoming these challenges

→ Tier-1 depth-based detection is independent of the learned model, so avoidance holds on unfamiliar terrain. Train on off-road datasets RUGD and RELIS-3D.

→ Stereo-inertial SLAM removes drift structurally – depth comes from geometry, not from a scale estimate that can drift.

→ Stereo depth is more robust to lighting shifts; segmentation trained on varied lighting; Tier-1 depth layer keeps working regardless.

Potential impact on the target audience

Cost-effective autonomy for the domains that need it most and can least afford expensive sensor suites

Search & rescue

Navigates where GPS is blocked; safer in disaster zones without constant remote monitoring

Precision agriculture

Camera-only autonomy under tree cover with a budget-friendly sensor set

Outdoor delivery robots

Low-cost last-mile robots where GPS is unreliable and LiDAR is too costly to scale

Benefits of the solution (social, economic, environmental, etc.)

Economic

Lower hardware cost – no LiDAR; cheaper camera and compute

Social / Safety

Safer operation in disaster and rescue zones without a human monitoring every move

Operational

Reduced need for manual monitoring

Scalable

Same architecture across outdoor domains – retrain the perception layer, no redesign

Environmental

Reduced resource footprint — camera-based sensing lowers hardware requirements and supports energy-efficient autonomous operation.

Details/Links of the reference and research work

Terrain datasets

RUGD

Purpose-built off-road terrain segmentation dataset

RELLIS-3D

Purpose-built off-road terrain segmentation dataset

Visual SLAM (localization)

ORB-SLAM3

Established stereo-inertial visual SLAM framework

VINS-Fusion

Established stereo-inertial visual SLAM framework

Planning & navigation

ROS 2 Nav2

Documented, widely deployed navigation stack across the robotics community

Real-time segmentation

Fast-SCNN

Architecture designed for real-time semantic segmentation

BiSeNetV2

Architecture designed for real-time semantic segmentation